2. If two elements have same first letter, then the first and second letters of the name are used as symbols. The first letter is in uppercase and the second letter is in lowercase.

Table 9.2 Elements with same first letter

Element	Symbol	Element	Symbol
Aluminium	Al	Hydrogen	H
Barium	Ba	Helium	He
Beryllium	Be	Nickel	Ni
Bismuth	Bi	Neon	Ne
Bromine	Br	Silicon	Si
Cobalt	Co	Sulphur	S

3. If the first two letters of the names of the elements are same, then the symbol consists of first letter and second or third letter of English name that they do not have in common.

Table 9.3 Elements with same two letters

Element	Symbol	Element	Symbol
Argon	Ar	Calcium	Ca
Arsenic	As	Cadmium	Cd
Chlorine	Cl	Magnesium	Mg
Chromium	Cr	Manganese	Mn

4. Some symbols are used on the basis of their Greek name or Latin name of the elements. There are eleven such elements.

Table 9.4 Greek or Latin name of elements

Element	Latin Name	Symbol
Sodium	Natrium	Na
Potassium	Kalium	K
Iron	Ferrum	Fe
Copper	Cuprum	Cu
Silver	Argentum	Ag
Gold	Aurum	Au
Mercury	Hydrargyrum	Hg
Lead	Plumbum	Pb
Tin	Stannum	Sn
Antimony	Stibium	Sb
Tungsten	Wolfram	W

5. Some elements are named using the name of the country / scientist / colour / mythological character / planet.

Table 9.5 Elements named using name of the country, scientist etc.

Name	Symbol	Name derived from
Americium	Am	**Am**erica (Country)
Europium	Eu	**Eu**rope (Country)
Nobelium	No	Alfred **No**bel (Scientist)
Iodine	I	Violet (Colour, Greek)
Mercury	Hg	God Mercury (Mythologic character)
Plutonium	Pu	Pluto (Planet)
Neptunium	Np	Neptune (Planet)
Uranium	U	Uranus (planet)

9.1.2 Writing the Symbols

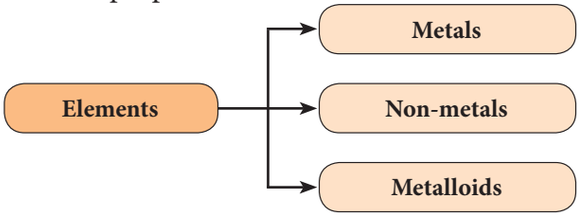
While writing the symbol for an element, we should adhere to the following rules.

1. If the element has a single English letter as a symbol, it should be written in capital letter.
2. For elements having two letter symbols, the first letter should be in capital followed by small letter

9.2 Metals and Non-metals

The progress of man towards civilization is linked with the discovery of several metals and non-metals. Even today, the index of prosperity of a country depends upon the amount of metals and non-metals it produces. The wealth of a country is measured by the amount of gold in its reserve.

An element can be identified as metal or non-metal by comparing its properties with the general properties of metals and non-metals. In doing so, we find that some elements neither fit with the metals nor with non-metals. Such elements are called semi-metals or metalloids. Elements are classified into metals, non-metals, and metalloids based on their properties.



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